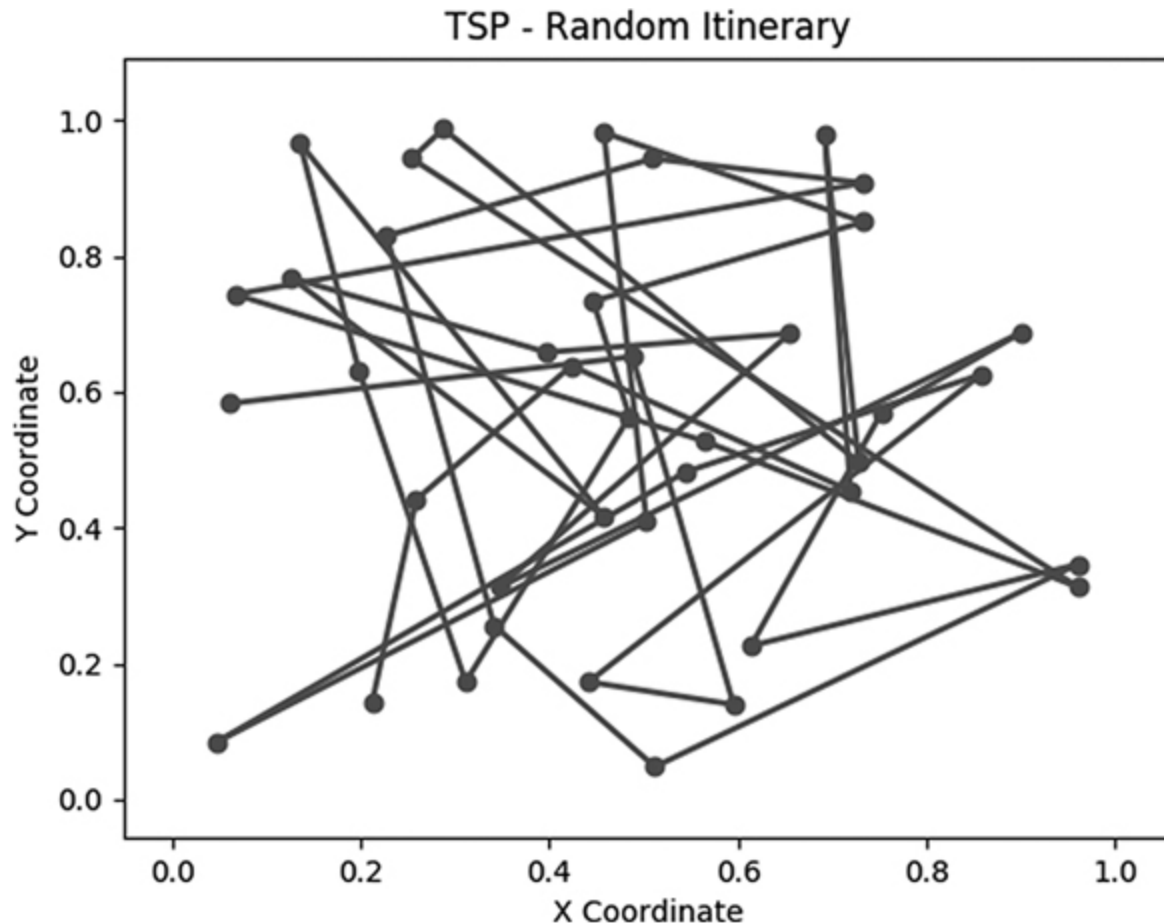


of lines. Then it plots the points of the itinerary and the lines we've created connecting them.

If you plot the itinerary with `plotitinerary(cities, itinerary, 'TSP - Random Itinerary', 'figure2')`, you'll generate the plot shown in [Figure 6-2](#).



[Figure 6-2](#): The itinerary resulting from visiting the cities in the random order in which they were generated

Maybe you can tell just by looking at [Figure 6-2](#) that we haven't yet found the best solution to the TSP. The itinerary we've given our poor salesman has him whizzing all the way across the map to an extremely distant city several times, when it seems obvious that he could do much better by stopping at some other cities along the way. The goal of the rest of this chapter is to use algorithms to find an itinerary with the minimum traveling distance.